Solution Requirements

Date	28 JUNE 2027
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users securely access the application to submit credit card details and view prediction results.	High
2	Authorization levels	Admin can manage the system and model, while users can only enter applicant details and view predictions.	High

3	External interfaces	*The system uses the Application Record and Credit Record datasets and interacts with the Flask web interface.*	Medium
4	Transactions processing	The system accepts applicant details, processes them using the trained machine learning model, and returns an approval or rejection prediction.	High
5	Reporting	The system displays the prediction result and can generate model performance metrics such as accuracy and confusion matrix.	Medium
6	Business rules, etc.	The prediction is based on applicant information such as income, employment, family status, age, and credit history using the trained ML model.	High
7	Compliance to laws or regulations	Applicant information should be handled securely and maintained with data privacy and confidentiality.	High
8	*Other*	The application should provide a simple, user-friendly interface and return prediction results quickly.	Medium

Step 2: Non-Functional Requirements (NFR)

S.NO	NFR Category	Requirement Description	Target Metric/ Acceptance Criteria
1	Performance & Speed	The system should process applicant details and display the prediction quickly.	Prediction result displayed within 2 seconds.
2	Scalability	The system should be able to handle multiple users submitting applications.	Supports 100+ users without significant delay.
3	Security & Data Privacy	Applicant information should be stored and processed securely to protect personal data.	Data is processed securely and user information is kept confidential.
4	Reliability & Availability	The application should be available whenever users need to access it and provide accurate predictions.	99% system availability during normal usage.
5	Usability & Accessibility	The application should have a simple, user-friendly interface that is easy to understand and use.	Users can submit an application and receive a prediction without assistance.
6	*Other* (Maintainability)	The system should be easy to update, maintain, and improve with new datasets or machine learning models.	New models or datasets can be integrated with minimal code changes.